

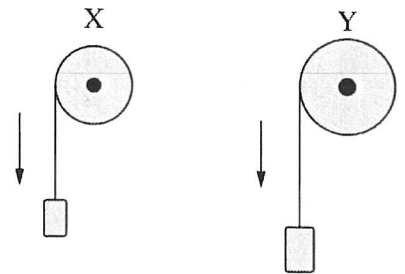
A Side

B

1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration a . As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity ω_Y of Pulley Y compare to the final angular velocity ω_X of Pulley X?

$$v_x = v_y$$

$$r_Y > r_X$$



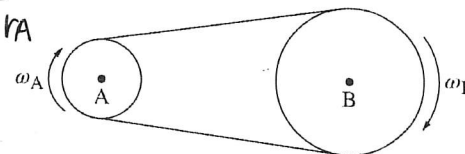
- (A) $\omega_Y > \omega_X$
 (B) $\omega_Y < \omega_X$
 (C) $\omega_Y = \omega_X$
 (D) It cannot be determined without knowing the relative masses of each block.

C

2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed ω_A and Wheel B has constant angular speed ω_B . Which of the following correctly relates ω_A

$$v_A = v_B$$

$$r_B = 3r_A$$



(A) $\omega_A = \frac{\omega_B}{3}$

$$v_A = \frac{v_B}{3}$$

(B) $\omega_A = \omega_B$

(C) $\omega_A = 3\omega_B$

(D) $\omega_A = 6\omega_B$

D

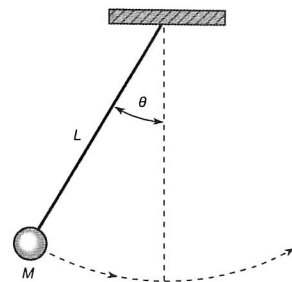
3. A pendulum is formed from a mass M connected to a string of length L . The pendulum is initially raised to form an angle θ (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be T (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?

(A) $\frac{\theta}{T}$

(B) $\frac{\theta L}{T}$

(C) $\frac{2\theta L}{T}$

(D) $\frac{4\theta L}{T}$

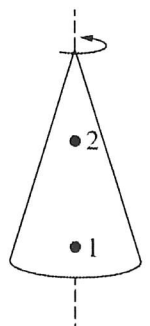
C

4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?

$$r_1 > r_2$$

$$v_1 > v_2$$

- (A) The angular speed of the insect is greater at Point 1 than at Point 2.
 (B) The angular speed of the insect is greater at Point 2 than at Point 1.
 (C) The linear speed of the insect is greater at Point 1 than at Point 2.
 (D) The linear speed of the insect is greater at Point 2 than at Point 1.



B Side

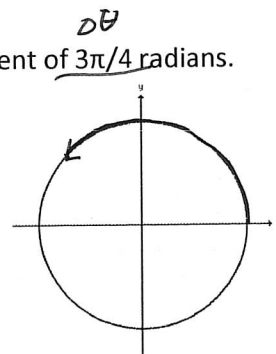
1. A disk has a radius of 2.5 m and undergoes an angular displacement of $3\pi/4$ radians.

- a. Sketch this angular displacement
b. Convert this measurement to degrees.

$$\frac{3}{4}\pi = 135^\circ$$

- c. Determine the arc length formed.

$$\frac{3}{4}\pi \times 2.5 = \frac{15}{8}\pi \text{ m}$$



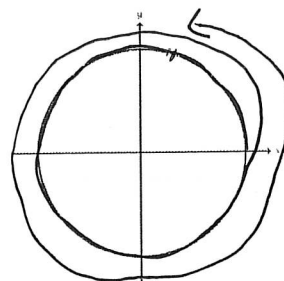
2. A woman walks in a circular path of radius 20 m. The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

- a. Sketch this angular displacement
b. Convert her angular displacement to radians.

$$2.2 \times 2\pi = 4.4\pi \text{ rad}$$

- c. Calculate the woman's linear displacement.

$$\Delta x = \Delta\theta \times r = 4.4\pi \times 20 = 88\pi \text{ m}$$



- d. Calculate the woman's angular velocity (in rad/s)

$$\omega = \frac{\Delta\theta}{t} = \frac{4.4\pi}{20} = 0.22\pi \text{ rad/s}$$

- e. Calculate the woman's linear velocity (in m/s)

$$v = \frac{\Delta x}{t} = \frac{88\pi}{20} = 4.4\pi$$

3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

- a. Determine the radius of the record

$$\Delta x = \Delta\theta \times r$$

$$r = \frac{\Delta x}{\Delta\theta} = \frac{1.3}{2\pi} = 0.65\pi \text{ m}$$

- b. Determine the angular velocity of the record (in deg/s)

$$\omega = \frac{\Delta\theta}{t} = \frac{2\pi}{1.2} = \cancel{1.67\pi} 300 \text{ deg/s}$$

